WHEN SPECIALIZATIONS LIFT, IMAGES OF SPECIALIZATION-STABLE SUBSETS STAY SPECIALIZATION-STABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter, staged here as sources/stacks-topology/preamble.tex; this card uses retained-preamble.tex, which copies that shared preamble mechanically, including its reference/slogan/history display environments. Nothing was added, removed or redefined; the wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: the chapter carries no chapter-local bibliography and neither the counted statement nor its proof contains a citation, so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: the card was taken from the staged candidate list for this chapter, candidate lemma-lift-specializations-images; its statement and proof spans are checked against the pinned chapter so that each span opens and closes a real, uncommented environment, and this lane confirms by rescanning that neither span carries a citation.

Allowed standard prerequisites (editorial).

- The specialization relation, stability under specialization and under generalization, and the lifting properties for a continuous map. Set up in the author's section on specialization, immediately before the counted statement. Fixes the two assertions and the hypotheses under which they hold.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces.*

(1) *If specializations lift along f, and if $T \subset X$ is stable under specialization, then $f(T) \subset Y$ is stable under specialization.*
(2) *If generalizations lift along f, and if $T \subset X$ is stable under generalization, then $f(T) \subset Y$ is stable under generalization.*

Proof. Let $y' \rightsquigarrow y$ be a specialization in Y where $y' \in f(T)$ and let $x' \in T$ such that $f(x') = y'$. Because specialization lift along f, there exists a specialization $x' \rightsquigarrow x$ of x' in X such that $f(x) = y$. But T is stable under specialization so $x \in T$ and then $y \in f(T)$. Therefore $f(T)$ is stable under specialization.

The proof of (2) is identical, using that generalizations lift along f. □

Date: stacks-project:topology.tex:local-snapshot-20261012; CC BY 4.0.